Preserve the Green & Golden Bell Frog:

A Journey to Save the Axolotls

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Overview

Batrachochytrium dendrobatidis, i.e. the Chytrid fungus, is an aggressive fungal disease threatening amphibian species worldwide. The fungus attaches to their skin, limiting the amphibian's ability to breathe, absorb water, and self-regulate. If untreated, Chytrid is fatal in most species. In Australia, the Green and Golden Bell Frog population has decreased by 90% due to this fungus. Researchers are actively working to preserve the population through a variety of treatment methods. We seek to find the optimal combination of treatments to preserve the population while minimizing cost.

The problem

We assume the setting of small, isolated populations of Green and Golden Bell Frogs in Australia. We want to optimize dynamic Chytrid treatment strategies for population retention and cost effectiveness.

Assumptions:

1. Frogs are Susceptible, Infected, Dead, or in sauna Treatment
2. SIR-like dynamics: closed population and mass action
3. No reproduction; frogs only die due to Chytrid infection
4. Money can be allocated to sauna (u_1) or antifungal (u_2) treatment
5. Some parameters are dependent on state and controls:
   a: u_1 determines carrying capacity K for T
   b: $\xi(u_2)$ determines the rate of transport from S to I
   c: Small I makes antifungal treatment more expensive per frog via ξ

Dynamics:

Resulting ODE dynamics (parameter dependencies suppressed):

$$\dot{S} = -\alpha IS + \gamma T + \eta\xi I - \zeta S\left(1 - \frac{T}{K}\right)$$
$$\dot{I} = \alpha IS - \zeta I\left(1 - \frac{T}{K}\right) - \eta\xi I - \beta I$$
$$\dot{T} = \zeta(I + S)\left(1 - \frac{T}{K}\right) - \gamma T$$
$$\dot{D} = \beta I$$

- u_1 represents frog sauna spend rate averaged over 7-year structure lifespan
- u_2 represents antifungal treatment spend rate
- γ, ζ, ξ, represent rates of frogs leaving the sauna, entering the sauna, and entering antifungal treatment, respectively
- Plausible initial parameters identified with help from Claude AI
- Manual parameter optimization to enforce expected dynamics

Control:

- Current attempts have used the autonomous model only
- Initial attempts to penalize dead frogs failed to converge. We settled on penalizing I as a good surrogate, as it determines the death rate
- We chose a quadratic cost functional for tractability. Quadratic financial costs also evoke diminishing returns, and quadratic infection costs hedge against population crash

$$J[\mathbf{x}, u_1, u_2] = \int_0^{t_f} cI^2 + u_1^2 + u_2^2 dt$$

We encountered more numerical problems after more experimentation. We responded by constraining sauna expenditure to be constant. This improved convergence, and is realistic since the saunas are semi-permanent structures. In tuning our models, we chose $c = 3$, which corresponds to the penalty for an infected frog. The choice of $c = 3$ was an arbitrary choice that gave reasonable results; this can be adjusted depending on financial constraints - higher values imply more utilization of financial resources and vice versa.

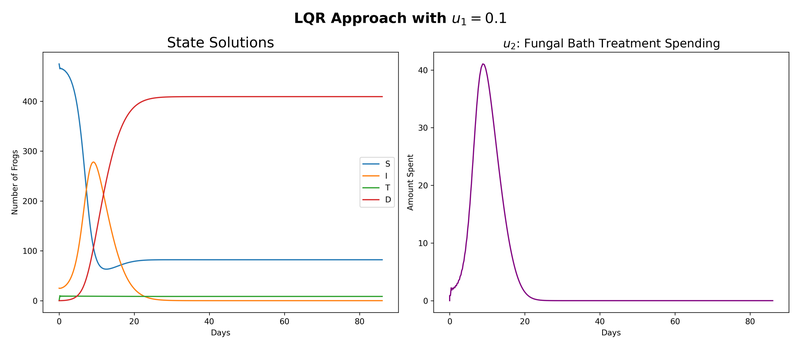
The Approach

Given our cost functional and the ODE system we'd defined, we decided to try solving for the optimal control u_2 in two ways:

- Linearize our ODE constraint at each time step and use infinite time LQR
- Use Hamilton's equations and scipy's `solve_bvp` package with desired initial and endpoint conditions

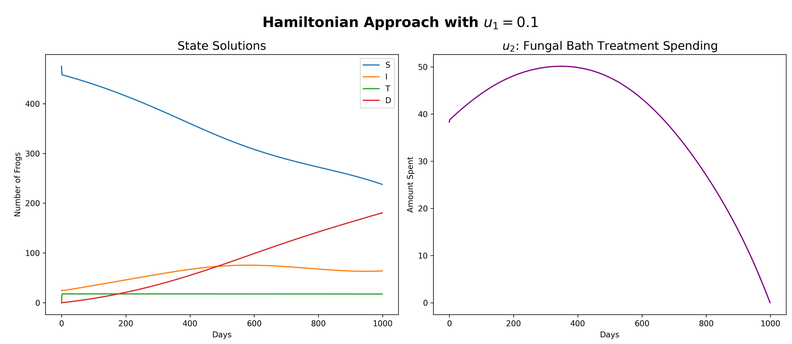
Infinite-Time LQR

Since the end conditions of the fungus spread are undetermined, we considered an iterated infinite-time horizon LQR approach. Because our system is also a function of our control, our initial linearization required an initial u_2 guess. We used an initial guess of $u_2 = 1$ and ran our solver with an initial state of 475 susceptible frogs, 25 infected, and a constant value of $u_1 = 0.1$. In our LQR results, the frogs die and dynamics stabilize more quickly than we'd expect.



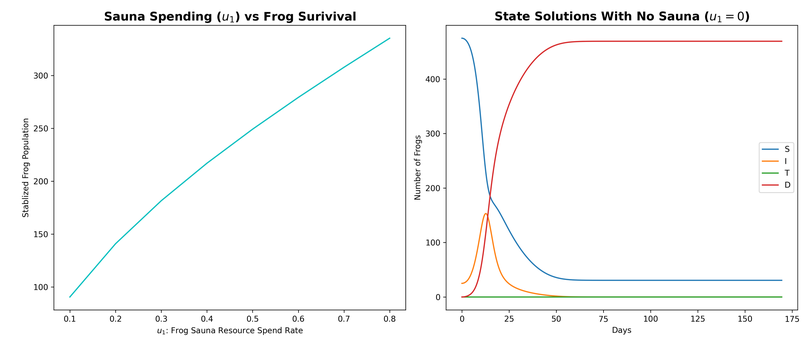
Hamiltonian

Another approach was to use the Hamiltonian and Pontryagin's maximum principle to solve for the optimal state and costate. Hamilton's equations give us differential equations for the state and costate and Pontryagin's maximum principle gives us a substitution for the control variable. Using these, we have a boundary value problem which can be solved using SciPy's `solve_bvp` module. As you can see from the figure below, when $u_1 = 0.1$, there are few frogs in the treatment class at the end of the 1000 days. Because u_1 controls the amount of space in the treatment class, increasing u_1 will allow for more frogs in the treatment class and helps preserve the population. This approach also showed considerable sensitivity to initial conditions.



Early Results & Analysis

We found that a model defined by an iterated infinite time horizon LQR approach produces relatively realistic dynamics of disease and treatment in the frog population. The LQR method shows that with constant nonzero sauna spending, our bath spending should be concentrated at the beginning, quickly spike before the peak of infection, then decrease gradually. Overall, we see that the optimal control under this model involves spending on both saunas and baths until the disease is under control, then relying on saunas to keep the population stable. Increasing the constant sauna spending parameter leads to quicker population stability and an increase in the number of surviving frogs. With zero sauna spending, the frog population nearly dies out. This indicates that saunas are necessary for keeping the frog population stable with our current c and are correlated with long term survival.



Using the Hamiltonian produced similar results: our u_2 solution correlates with I, suggesting more spending on antifungal treatment when infection is high to control spread, but ultimately, without the sauna treatment option, no optimal amount of spending on u_2 with our current c preserved the frog population.

Conclusions and next steps

Conclusions

Both approaches gave plausible solutions with some limitations. The direct Hamiltonian method exhibited considerable sensitivity to initial conditions, but gave more drawn-out, realistic dynamics, whereas our LQR model was more robust across different initial conditions, but produced rapid, less realistic dynamics. Despite these limitations, both models confirm that spending on antifungal treatment is crucial in controlling infection spikes.

Next Steps

- Find optimal control for the seasonal model, as the fungus is much more aggressive during colder months
- Incorporate a newly developed vaccine treatment, adding an immune class
- Non-vaccinated frogs are infection resistant after sauna treatment, implying flow from T to the new immune class
- Add a spontaneous infection rate, since our current model assumes that once all infected frogs are dead or have recovered, no more frogs will contract the fungus
- Vary vulnerability depending on frog stage of development; tadpoles and juveniles are significantly more susceptible
- Incorporate a natural reproduction and death rate, independent of the fungus
- Investigate differences between solve_bvp and ILQR
- Investigate ranges of c and u_1 and other aspects of current models

For references, see QR code in the upper left corner